Branched simplicial fans satisfy the gradient–Laplacian identity

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Abstract

The complex with facets $012, 013, 014$ is not a pseudomanifold, but its top boundary Laplacian and discrete-gradient generating polynomial both equal $\lambda^4(\lambda+2)^2(\lambda+5)$. This disproves the characterization in Conjecture 1 of the current Contreras–Tawfeek preprint. We give a uniform formula for any number of simplices joined along a common facet. The identified journal final remains unread, so campaign eligibility is pending.

Task 1. Full counterexample to the preprint formulation

Follow the source's conventions [1, pp. 3,8,18]: $B = \partial_d$ has rows indexed by the $(d-1)$-faces, columns indexed by the d-faces, and $\Delta_d = BB^{\mathsf{T}}$. Put $m = |K_d|$ and $n = |K_{d-1}|$. The set $\mathcal{M}_{m-i}(K)$ consists of discrete gradients with i matched pairs, all between these two dimensions; every cell of dimension below $d-1$ is critical. Thus it is the set of acyclic matchings of size i in the top two levels of the Hasse diagram. As in the source, $\mathcal{M}_j$ is empty for $j < 0$. Conjecture 1 asserts that

$$\det(\Delta_d + \lambda I_n) = \sum_{i=0}^{n} |\mathcal{M}_{m-i}(K)| \lambda^{n-i} \tag{1}$$

holds if and only if K is an orientable pseudomanifold.

Theorem 1. *Let $d \geq 2$, $r \geq 3$, and let S consist of d vertices. Choose distinct further vertices $u_1, \dots, u_r$, and let $K(d, r)$ be the simplicial complex generated by*

$$S \cup \{u_1\}, \dots, S \cup \{u_r\}.$$

This is a finite, connected, pure d-dimensional complex whose facet-adjacency graph is connected. It is not a pseudomanifold, and nevertheless it satisfies (1). Both sides are

$$\lambda^{1+r(d-1)}(\lambda + d)^{r-1}(\lambda + d + r). \tag{2}$$

Proof. Any two top faces meet in the common $(d-1)$-face S. Each top face has d other boundary faces, and each of these belongs to that top face alone. Hence $m = r$ and $n = 1 + rd$. In particular, S belongs to $r \geq 3$ top faces, violating the source's requirement of exactly two. It also violates the at-most-two condition if pseudomanifolds with boundary are allowed. The obstruction is therefore branching, rather than just the presence of a boundary or a disconnected complex.

Orient the top faces so that their incidence with S is $+1$. The supports of any two columns of B intersect only in that row. Every column has $d+1$ entries of absolute value 1. Consequently

$$B^{\mathsf{T}}B = dI_r + J_r,$$

where J_r is the all-ones matrix. Its eigenvalues are $d+r$ once and d with multiplicity $r-1$. The nonzero eigenvalues of BB^{T} are the same, and its nullity is $n-r$. This proves (2) for the determinant.

The bipartite incidence graph between the top two levels is a tree: one central vertex S is joined to r top-face vertices, each with d additional leaves. Every matching is acyclic, since a directed cycle would give an undirected cycle in this tree. The remaining Hasse edges point downward and cannot belong to a directed cycle, since no lower cell is matched. Thus every such matching is a discrete gradient of the required type. For completeness, this last assertion can be seen without an existence theorem: assign strictly decreasing positive values along the arrows in an acyclic orientation of these two levels, and assign smaller, dimension-increasing values to the lower levels. The resulting function has exactly the matched pairs as its reversed inequalities and satisfies the defining discrete Morse inequalities.

Choose the i top faces that will be matched. If the central face S is unused, there are d^i choices. If it is used, there are i choices for its partner and d^{i-1} choices for the others. Therefore, for $1 \le i \le r$,

$$|\mathcal{M}_{r-i}(K(d,r))| = \binom{r}{i}(d^i + id^{i-1}), \tag{3}$$

and the empty matching gives $|\mathcal{M}_r| = 1$. The binomial theorem and its derivative now give

$$\begin{aligned}\sum_{i=0}^{n} |\mathcal{M}_{r-i}|\lambda^{n-i} &= \lambda^{n-r}\big((\lambda+d)^r + r(\lambda+d)^{r-1}\big) \\ &= \lambda^{n-r}(\lambda+d)^{r-1}(\lambda+d+r).\end{aligned}$$

This is (2), completing the proof. □

The explicit five-vertex example

Take $S = \{0,1\}$ and $u_1 = 2, u_2 = 3, u_3 = 4$. For rows $01, 02, 03, 04, 12, 13, 14$ and columns $012, 013, 014$, the usual increasing-vertex orientations give

$$B = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad (|\mathcal{M}_3|, |\mathcal{M}_2|, |\mathcal{M}_1|, |\mathcal{M}_0|) = (1, 9, 24, 20).$$

Both polynomials are

$$\lambda^7 + 9\lambda^6 + 24\lambda^5 + 20\lambda^4.$$

The equivalence is therefore false. This counterexample does not refute its separate forward implication for orientable pseudomanifolds.

Why the identity also holds for incidence forests

More generally, (1) holds whenever the incidence graph of the top two levels is a forest. Here is a direct matrix explanation. Expansion in principal minors followed by Cauchy–Binet gives

$$\det(\lambda I_n + BB^{\mathsf{T}}) = \sum_{i=0}^{\min(m,n)} \lambda^{n-i} \sum_{\substack{U\subseteq[n],\ V\subseteq[m]\\ |U|=|V|=i}} \det(B_{U,V})^2. \tag{4}$$

Two different perfect matchings on the same selected vertices would have an alternating cycle in their union. Thus an incidence forest has at most one such matching. Each nonzero determinant in (4) is consequently 1 or -1, and its square counts that unique matching. All matchings are acyclic, so (4) is exactly (1). This is an elementary application of Cauchy–Binet and matching enumeration, not a claim that those classical tools are new.

Tasks 2–5. Proposed follow-up problems

2. Classify the complexes satisfying (1) when their top incidence graph is connected and has exactly one cycle, in terms of its signed incidences.

3. Determine whether (1) admits a pure, strongly connected two-dimensional counterexample to the characterization in which every edge belongs to at least two triangles.

4. Determine which stellar subdivisions preserve (1), starting with subdivisions of the common face in $K(d, r)$.

5. For fixed d and r, determine the largest number of top-dimensional discrete gradients among pure connected d-complexes with r facets and a forest as their top incidence graph.

Verification and review. The accompanying exact-arithmetic script constructs boundary matrices from simplex lists, enumerates matchings and checks their directed acyclicity, and compares their counts with the characteristic polynomial. The proof above establishes the whole family. All review was by the authoring agent; no independent peer review or proof-assistant certification is claimed.

References

[1] I. Contreras and A. R. Tawfeek, *On discrete gradient vector fields and Laplacians of simplicial complexes*, arXiv:2105.05388v2, 2 July 2022, 20 pages; Conjecture 1, p. 18. An identified journal version appeared in *Annals of Combinatorics* **28** (2024), 67–91, doi:10.1007/s00026-023-00655-1. The journal version has not been read in this audit.